16.1 The First Law of Thermodynamics

Question Paper

Course	CIE A Level Physics
Section	16. Thermodynamics
Topic	16.1 The First Law of Thermodynamics
Difficulty	Medium

Time allowed: 50

Score: /41

Percentage: /100

Question 1a

A fixed mass of an ideal gas has a pressure *p* and a volume *V*.

The gas undergoes a cycle of changes, A to B to C, as shown in Fig. 1.1

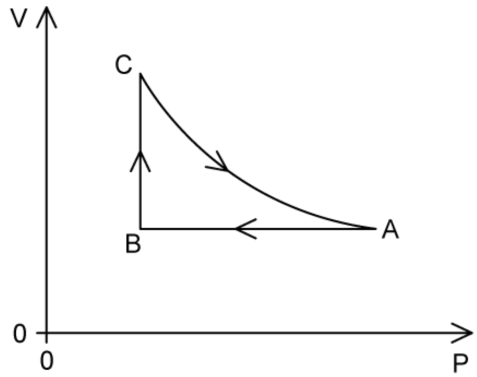


Fig. 1.1

Table 1.1 shows data for *p*, *V* and *T* for points A, B and C.

Table 1.1

	p / 10^5 Pa	V / 10^{-3} m^3	T / K
A			680
B	1.6	4.4	451
C			698

State the change in internal energy ΔU for one complete cycle, ABCA.

ΔU .... J

[1 mark]

Question 1b

Calculate the number of particles *N* in the gas.

[2 marks]

Question 1c

Complete Table 1.1 by filling in the missing pressures and volumes at A, B and C.

[2 marks]

Question 1d

(i)

The first law of thermodynamics of a system can be represented by the equation:

$$\Delta U = q + W$$

State, with reference to the system, what is meant by ΔU, q and W

[3]

(ii)

Explain how the first law of thermodynamics applies to the change C to A.

[2]

[5 marks]

Question 2a

A fixed mass of an ideal gas is initially at a temperature of 20 °C.

The gas has a volume of 0.12 m^3 and a pressure of 0.6×10^5 Pa.

State what is meant by an ideal gas.

[2 marks]

Question 2b

The gas undergoes three successive changes, as shown in Fig. 1.1.

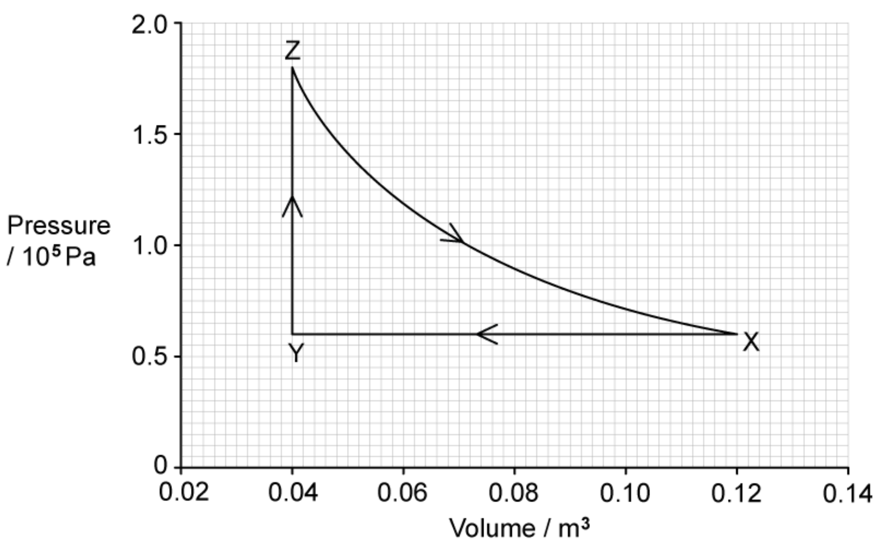


Fig. 1.1

The initial state is represented by point X. The gas is cooled at constant pressure to point X by the removal of 24.0 kJ of thermal energy.

The gas is then heated at constant volume to point Z.

FInally, the gas expands at constant temperature back to its original pressure and volume at point Z. During the expansion, the gas does 15.8 kJ of work.

Calculate the magnitude of the work done in kJ during the change XY.

[2 marks]

Question 2c

Complete Table 1.1 to show the work done on the gas, the thermal energy supplied to the gas and the increase in internal energy of the gas, for each of the changes XY, YZ and ZX.

Table 1.1

change	increase in internal energy of gas / kJ	thermal energy supplied to gas / kJ	work done on gas / kJ
XY		–24.0	
YZ			
ZX			–15.8

[3 marks]

Question 3a

Define the work done by a gas.

[2 marks]

Question 3b

A mass of water with negligible volume is vapourised into water vapour of volume $0.82\ m^3$ at a temperature of 100 °C and atmospheric pressure 1.0×10^5 Pa. The thermal energy supplied to the water is 9.5×10^5 J.

Calculate the magnitude of the work done by the water in expanding against the atmosphere when it vapourises.

work done .... J

[2 marks]

Question 3c

Determine the increase in internal energy of the water when it vapourises at 100 °C. Explain your reasoning.

increase in internal energy = ... J

[2 marks]

Question 3d

Suggest, with a reason, how the specific latent heat of vapourisation of water at a pressure greater than atmospheric pressure compares with its value at atmospheric pressure.

[3 marks]

Question 4a

State the first law of thermodynamics and include any relevant units.

[2 marks]

Question 4b

A fixed mass of water is in a beaker at atmospheric pressure. The initial temperature of the water is 0 °C. The water is supplied with thermal energy q, so that its temperature increases to 10 °C. There is no net change in the volume of the water.

Determine the work done and the increase in internal energy of the water by completing **Table 1.1**, considering the first law of thermodynamics.

Table 1.1

Work done on water	thermal energy supplied to water	increase in internal energy of water
	$+q$	

[2 marks]

Question 4c

The water from part (b) is now heated so it's temperature increases by a further 10 °C to a final temperature of 20 °C. This process causes the volume of the water to increase so that work *W* is done.

Assume that the change in internal energy is the same as in part (b).

Determine the work done, thermal energy supplied and the increase in internal energy of the water by completing **Table 1.2**, considering the first law of thermodynamics.

Table 1.2

Work done on water	thermal energy supplied to water	increase in internal energy of water

[2 marks]

Question 4d

Explain how the average specific heat capacity of the water, in parts (b) and (c), between 10 °C and 20 °C compares with its average value between 0 °C and 10 °C.

[2 marks]

Question 5a

Define internal energy.

[1 mark]

Question 5b

State and explain, in molecular terms, whether the internal energy of the following increases, decreases or does not change.

(i)
A lump of copper as it is cooled.

[3]

(ii)
Some water as it evaporates at a constant temperature.

[3]

[6 marks]

Model Answers: Medium

1a

(a) The change in internal energy is:

- 0 J [1 mark]

[Total: 1 mark]

This is a 1 mark question, so you can see that **no calculation** is required. The **cycle of changes** goes from A to B to C and then **back to A**.

1b

(b) Calculate the number of particles *N* in the gas:

List the known quantities:

- Pressure, $p = 1.6 \times 10^5$ Pa
- Volume, $V = 4.4 \times 10^{-3}$ m^3
- Temperature, $T = 451$ K
- Boltzmann constant, $k = 1.38 \times 10^{-23}$ J K^{-1}

Use the ideal gas equation:

- $pV = NkT$
- $(1.6 \times 10^5) \times (4.4 \times 10^{-3}) = N \times (1.38 \times 10^{-23}) \times 451$
- $\frac{(1.6 \times 10^5) \times (4.4 \times 10^{-3})}{(1.38 \times 10^{-23}) \times 451} = N$ [1 mark]
- $N = 1.13 \times 10^{23}$ [1 mark]

[Total: 2 marks]

1c

(c) Complete table 1.1:

List the known quantities:

- Number of particles in gas, $N = 1.13 \times 10^{23}$

A completed table that would score 2 marks should look like this:

	p / 10^5 Pa	V / 10^{-3} m^3	T / K
A	**2.4**	**4.4**	680
B	1.6	4.4	451
C	**1.6**	**6.8**	698

- Volume for A = 4.4 read from the graph and pressure for C = 1.6 read from the graph; [1 mark]
- Pressure for A = 2.4 and volume for B = 6.8 calculated from $pV = NkT$ [1 mark]

[Total: 2 marks]

Using the **diagram** it is possible to see that the volume at A and B is the same.

To calculate the missing volume for C it is possible to use the equation ***pV = NkT*** from (b) but this takes **longer** than looking at the relationship $pV \propto T$.

$$\frac{T_A}{T_C} = \frac{682}{440} = 1.55$$

$$V_c = 1.55V_A = 1.55 \times 4.4 = 6.8$$

1d

(d)

(i) The meaning of the terms in the equation of the first law of thermodynamics are:

- ΔU is the increase in the internal energy (of the system); [1 mark]
- q is the thermal energy supplied to the system; [1 mark]
- W is the work done on the system; [1 mark]

(ii) The first law of thermodynamics applies to the change C to A because:

- Volume decreases and work is done on the gas; [1 mark]
- Temperature decreases and internal energy decreases; [1 mark]

[Total: 5 marks]

2a

(a) An ideal gas is:

- A gas that obeys (the relationship) $pV \propto T$ [1 mark]
- Where p = pressure of the gas, V = volume of the gas and T = thermodynamic temperature of the gas; [1 mark]

[Total: 2 marks]

2b

(b) Calculate the magnitude of the work done in kJ during the change XY

List the known quantities:

- For X and Y pressure, $p = 0.6 \times 10^6$ Pa
- For X, volume, $V_X = 0.12$ m^3
- For Y, volume, $V_Y = 0.04$ m^3

Use the work done on a gas equation:

- $W = p \Delta V$ [1 mark]
- $W = p\left(V_X - V_Y\right)$
- $W = (0.6 \times 10^6) \times (0.12 - 0.04)$
- $W = 48\,000 \text{ J} = 48 \text{ kJ}$ [1 mark]

[Total: 2 marks]

2c

(c) A correctly completed table should look like this:

change	increase in internal energy of gas / kJ	thermal energy supplied to gas / kJ	work done on gas / kJ
XY	**24.0**	–24.0	**48.0**
YZ	**–24.0**	**24.0**	**0**
ZX	**0**	**15.8**	–15.8

- *1 mark for each correct row*

[Total: 3 marks]

During XY

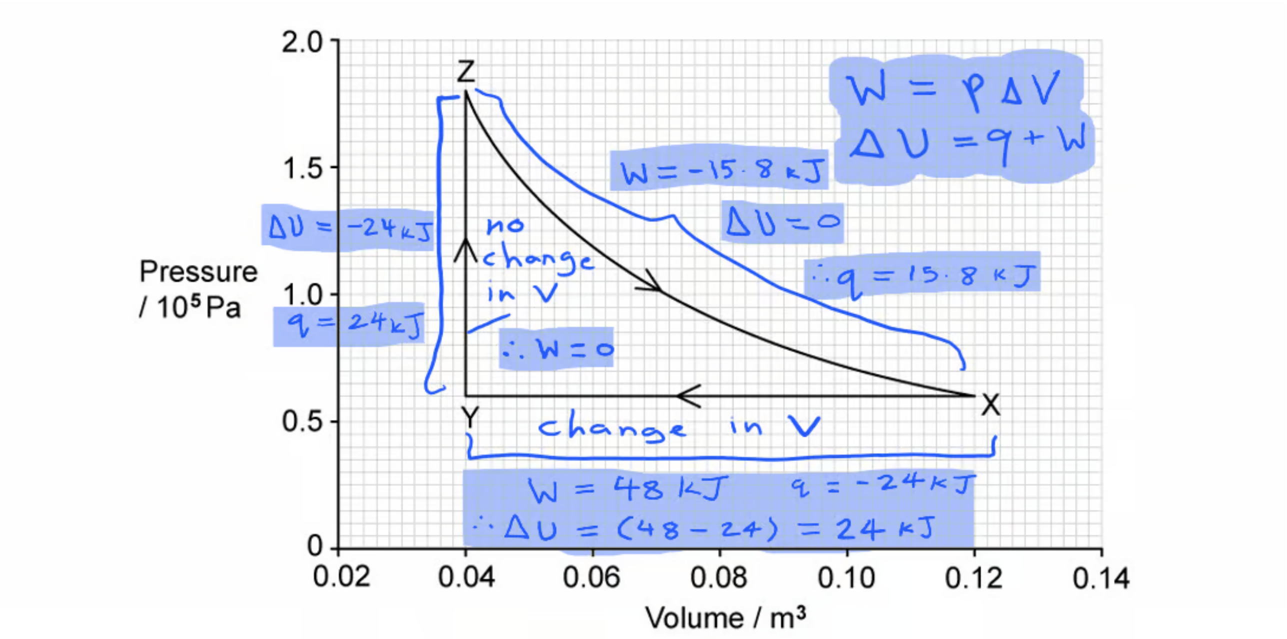
- Work done for XY = 48.0 kJ (from **(c)**)
- Increase in internal energy ΔU = 48 – 24 = 24.0 kJ calculated from the equation $\Delta U = q + W$

During YZ

- Work done for YZ = 0 (as there is no change in volume)
- Increase in internal energy ΔU = –24.0 kJ because the total increase in internal energy of the system (XYZX) = 0
- Hence, thermal energy = 24.0 kJ

During ZX

- Work done for ZX = –15.8 kJ (given in table)
- Increase in internal energy = 0 (as volume increases)
- Hence, thermal energy = 15.8 kJ



3a

(a) The work done by a gas is defined as:

- The work done by a volume of gas that changes at constant pressure; [1 mark]
- $W = p\,\Delta V$; [1 mark]

[Total: 2 marks]

3b

(b) Calculate the magnitude of the work done by the water in expanding against the atmosphere when it vapourises:

List the known quantities:

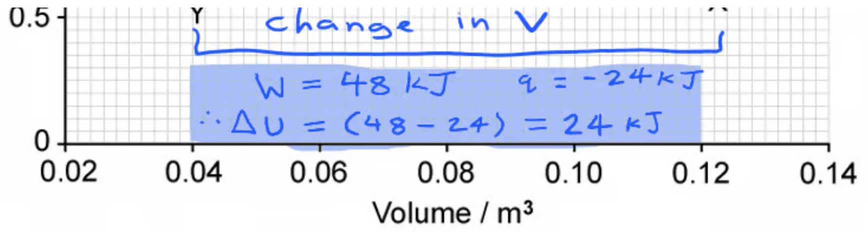
- Atmospheric pressure, $p = 1.0 \times 10^5$ Pa
- Volume of water vapour, $V_f = 0.82$ m^3
- Volume of water, $V_i = 0$ m^3

Use the work done by a gas equation:

- $W = p\,\Delta V$ [1 mark]
- $W = (1.0 \times 10^5) \times 0.82$
- $W = 82\,000 \text{ J} = 8.2 \times 10^4 \text{ J}$ [1 mark]

[Total: 2 marks]

3c



3a

(a) The work done by a gas is defined as:

- The work done by a volume of gas that changes at constant pressure; [1 mark]
- $W = p\,\Delta V$; [1 mark]

[Total: 2 marks]

3b

(b) Calculate the magnitude of the work done by the water in expanding against the atmosphere when it vapourises:

List the known quantities:

- Atmospheric pressure, $p = 1.0 \times 10^5$ Pa
- Volume of water vapour, $V_f = 0.82$ m^3
- Volume of water, $V_i = 0$ m^3

Use the work done by a gas equation:

- $W = p\,\Delta V$ [1 mark]
- $W = (1.0 \times 10^5) \times 0.82$
- $W = 82\,000 \text{ J} = 8.2 \times 10^4 \text{ J}$ [1 mark]

[Total: 2 marks]

3c

(c) Determine the increase in internal energy of the water when it vapourises at 100 °C:

List the known quantities:

- Work done by the water vapour, W = 82 000 J = 0.82 × 10^5 J (from part (b))
- Thermal energy supplied to the water, q = 9.5 × 10^5 J (from part (b))

Determine the sign of the work done by the gas:

- (Water expands and does work against the atmosphere so) the work done on the water is negative; [1 mark]

Use the first law of thermodynamics:

- $\Delta U = q + W$
- $\Delta U = (9.5 - 0.82) \times 10^5 = 8.68 \times 10^5$ J [1 mark]

[Total: 2 marks]

3d

(d) When the atmospheric pressure is greater then the specific latent heat of vapourisation will be as follows:

- The greater the atmospheric pressure, the higher the boiling point; [1 mark]
- So a larger amount of thermal energy is needed to overcome the intermolecular forces (between the water molecules); [1 mark]

Consider the effect on the specific latent heat of vapourisation:

- Thermal energy, $q = mL_v$
- Hence, specific latent heat of vapourisation also increases; [1 mark]

[Total: 3 marks]

This is a **difficult question**. The **atmospheric pressure** above the water is **higher**, so the water requires **more energy** to overcome this.

As **atmospheric pressure decreases**, you walk **higher up** a mountain, then the **boiling temperature** of your **water decreases**, so the time it takes your water to boil is less and the **energy required to boil** is also **less**. This can be an easier way to think about this question.

4a

(a) The first law of thermodynamics states:

- Increase in internal energy, ΔU = energy supplied to the system by heating, q + work done on the system, W; [1 mark]
- Units are J; [1 mark]

[Total: 2 marks]

4b

(b) A table that would score 2 marks should look like this:

work done on water	thermal energy supplied to water	increase in internal energy of water
0	$+q$	***+ΔU***

- The volume of the water does not change, so the work done on the water = 0; [1 mark]
- The water remains at atmospheric pressure, and the only energy supplied is thermal energy, so the internal energy of the water increases by $+\Delta U$; [1 mark]

[Total: 2 marks]

4c

(c) A table that would score 2 marks should look like this:

Work done on water	thermal energy supplied to water	increase in internal energy of water
–W	***+W +ΔU***	***+ΔU***

- Work, W is done on the water to increase the volume, so the work done by the water = $-W$

AND

- The change in internal energy = same as part (b) = $+\Delta U$; [1 mark]

- Work done + thermal energy supplied = increase in internal energy
  - So $-W$ + thermal energy supplied = $+\Delta U$
  - Hence, thermal energy supplied = $+W + \Delta U$; [1 mark]

[Total: 2 marks]

4d

(d) The average specific heat capacity of water between 10 °C and 20 °C compares with its average value between 0 °C and 10 °C:

- The change in thermal energy is higher; [1 mark]
- So the specific heat capacity is higher; [1 mark]

[Total: 2 marks]

The change in thermal energy from 0 °C to 10 °C = $+q$

The change in thermal energy from 10 °C to 20 °C = $+W + \Delta U$

Hence, $+W + \Delta U > +q$

It can be easier to consider these values using the specific heat capacity equation:

$$\Delta Q = mc\Delta\theta$$

So specific heat capacity, $c = \dfrac{\Delta Q}{m\Delta\theta}$

5a

(a) Internal energy is defined as

- The sum of the random distribution of kinetic and potential energies within a system of molecules; [1 mark]

[Total: 1 mark]

5b

(b)

(i) The internal energy of a lump of copper as it cools:

- The potential energy is unchanged as the atoms remain in the same positions (in the solid); [1 mark]
- The vibrational kinetic energy reduces because the temperature gets lower; [1 mark]
- So the internal energy is less; [1 mark]

(ii) The internal energy of some water as it evaporates at a constant temperature:

- The potential energy increases because the separation of the atoms increases (as the water molecules move from a liquid to evaporate as a gas); [1 mark]
- The kinetic energy remains constant because the temperature remains constant; [1 mark]
- So the internal energy increases; [1 mark]

[Total: 6 marks]